

Mechanisms of microglia-mediated synapse turnover and synaptogenesis.

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Microglia shape synaptic connections among neurons of the central nervous system (CNS) during development and adulthood. In this review, the actions by which they facilitate pruning, refinement, and new synaptic development throughout the lifespan are considered, along with the molecular mechanisms by which neurons and microglia communicate to guide these actions. Microglia survey neuronal activity and selectively modify synaptic connections at the level of individual dendrites and synapses. This is important given that microglia are necessary for a healthy nervous system capable of learning and other neural phenomena based on synaptic modifications and can also cause pathological synaptic disfunctions in immunologically driven neurodegenerative diseases. Understanding how microglia directly shape synaptic connections between neurons yields a more complete understanding of normal neuroplasticity and provides new routes for understanding disease states.

Microglia Are Dispensable for Developmental Dendrite Pruning of Mitral Cells in Mice.

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During early development, neurons in the brain often form excess synaptic connections. Later, they strengthen some connections while eliminating others to build functional neuronal circuits. In the olfactory bulb, a mitral cell initially extends multiple dendrites to multiple glomeruli but eventually forms a single primary dendrite through the activity-dependent dendrite pruning process. Recent studies have reported that microglia facilitate synapse pruning during the circuit remodeling in some systems. It has remained unclear whether microglia are involved in the activity-dependent dendrite pruning in the developing brains. Here, we examined whether microglia are required for the developmental dendrite pruning of mitral cells in mice. To deplete microglia in the fetal brain, we treated mice with a colony-stimulating factor 1 receptor (CSF1R) inhibitor, PLX5622, from pregnancy. Microglia were reduced by >90% in mice treated with PLX5622. However, dendrite pruning of mitral cells was not significantly affected. Moreover, we found no significant differences in the number, density, and size of excitatory synapses formed in mitral cell dendrites. We also found no evidence for the role of microglia in the activity-dependent dendrite remodeling of layer 4 (L4) neurons in the barrel cortex. In contrast, the density of excitatory synapses (dendritic spines) in granule cells in the olfactory bulb was significantly increased in mice treated with PLX5622 at postnatal day (P) 6, suggesting a role for the regulation of dendritic spines. Our results indicate that microglia do not play a critical role in activity-dependent dendrite pruning at the neurite level during early postnatal development in mice.

Microglia in the pathogenesis of autism spectrum disorders.

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Proper synaptic pruning is essential for the development of functional neural circuits. Impairments in synaptic pruning disrupt the excitatory versus inhibitory balance (E/I balance) of synapses, which may cause neurodevelopmental disorders such as autism spectrum disorder (ASD). Recent studies have determined molecular mechanisms by which microglia, the brain's resident immune cells, engulf inappropriate and less active synapses. Thus, microglial dysfunction may be involved in the pathogenesis

of ASD through attenuated or excess synaptic pruning. In this review, we discuss recent animal and human studies that report an E/I imbalance and the characteristics of microglia in ASD. We will further discuss whether and how synaptic pruning by microglia is involved in the pathogenesis of ASD.

Synaptic pruning of murine adult-born neurons by microglia depends on phosphatidylserine.

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New neurons, continuously added in the adult olfactory bulb (OB) and hippocampus, are involved in information processing in neural circuits. Here, we show that synaptic pruning of adult-born neurons by microglia depends on phosphatidylserine (PS), whose exposure on dendritic spines is inversely correlated with their input activity. To study the role of PS in spine pruning by microglia in vivo, we developed an inducible transgenic mouse line, in which the exposed PS is masked by a dominant-negative form of milk fat globule-EGF-factor 8 (MFG-E8), MFG-E8D89E. In this transgenic mouse, the spine pruning of adult-born neurons by microglia is impaired in the OB and hippocampus. Furthermore, the electrophysiological properties of these adult-born neurons are altered in MFG-E8D89E mice. These data suggest that PS is involved in the microglial spine pruning and the functional maturation of adult-born neurons. The MFG-E8D89E-based genetic approach shown in this study has broad applications for understanding the biology of PS-mediated phagocytosis in vivo.

CRISPRi-based screen of Autism Spectrum Disorder risk genes in microglia uncovers roles of ADNP in microglia endocytosis and synaptic pruning.

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Autism Spectrum Disorders (ASD) are a set of neurodevelopmental disorders with complex biology. The identification of ASD risk genes from exome-wide association studies and de novo variation analyses has enabled mechanistic investigations into how ASD-risk genes alter development. Most functional genomics studies have focused on the role of these genes in neurons and neural progenitor cells. However, roles for ASD risk genes in other cell types are largely uncharacterized. There is evidence from postmortem tissue that microglia, the resident immune cells of the brain, appear activated in ASD. Here, we used CRISPRi-based functional genomics to systematically assess the impact of ASD risk gene knockdown on microglia activation and phagocytosis. We developed an iPSC-derived microglia-neuron coculture system and high-throughput flow cytometry readout for synaptic pruning to enable parallel CRISPRi-based screening of phagocytosis of beads, synaptosomes, and synaptic pruning. Our screen identified ADNP, a high-confidence ASD risk genes, as a modifier of microglial synaptic pruning. We found that microglia with ADNP loss have altered endocytic trafficking, remodeled proteomes, and increased motility in coculture.

New Insights Regarding Diagnosis and Medication for Schizophrenia Based on Neuronal Synapse-Microglia Interaction.

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Schizophrenia is a common psychiatric disorder that usually develops during adolescence and young adulthood. Since genetic and environmental factors are involved in the disease, the molecular status of the pathology of schizophrenia differs across patients. Recent genetic studies have focused on the association between schizophrenia and the immune system, especially microglia-synapse interactions. Microglia physiologically eliminate unnecessary synapses during the developmental period. The overactivation of synaptic pruning by microglia is involved in the pathology of brain disease. This paper focuses on the synaptic pruning function and its molecular machinery and introduces the hypothesis that excessive synaptic pruning plays a role in the development of schizophrenia. Finally, we suggest a strategy for diagnosis and medication based on modulation of the interaction between microglia and synapses. This review provides updated information on the involvement of the immune system in schizophrenia and proposes novel insights regarding diagnostic and therapeutic strategies for this disease.

Dysfunctional synaptic pruning by microglia correlates with cognitive impairment in sleep-deprived mice: Involvement of CX3CR1 signaling.

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Microglia are involved in sleep/wake cycles and the response to sleep loss. Synaptic pruning by microglia is necessary for central nervous system circuit refinement and contributes to cognitive function. Here, we investigated whether and how microglia-mediated synaptic pruning may be involved in cognitive deficits induced by sleep deprivation in mice. Mice were deprived of sleep by leaving them in a spontaneously rotating rod for 72 h, after which their cognitive function was assessed using an object location test, Y maze, and novel object recognition test. Sleep deprivation lowered the discrimination index for familiar locations in the object location test and Y maze. Microglial morphology was assessed using immunostaining Iba1, while microglia-mediated synaptic pruning was examined based on immunostaining PSD95, CD68, and Iba1. Sleep deprivation also activated microglial cells in the hippocampus, as reflected in bigger soma as well as fewer and shorter branches than normal sleep. Sleep deprivation downregulated phagocytic markers and internalization of postsynaptic protein 95 (PSD95), suggesting impaired synaptic pruning. CX3C motif chemokine receptor 1 (CX3CR1) signaling was detected in in vitro experiments. Sleep deprivation also downregulated CX3CR1. Activation of CX3CR1 signaling increased phagocytosis activity of BV2 microglia in vitro. Sleep deprivation dysregulates microglial CX3CR1 signaling and inhibits synaptic pruning, contributing to associated cognitive deficits. These findings identify CX3CR1-dependent synaptic pruning as a potential therapeutic target in which sleep deprivation causes recognition impairments.

Deficient autophagy in microglia impairs synaptic pruning and causes social behavioral defects.

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Autism spectrum disorders (ASDs) are neurodevelopmental disorders caused by various genetic and environmental factors that result in synaptic abnormalities. ASD development is suggested to involve microglia, which have a role in synaptic refinement during development. Autophagy and related

pathways are also suggested to be involved in ASDs. However, the precise roles of microglial autophagy in synapses and ASDs are unknown. Here, we show that microglial autophagy is involved in synaptic refinement and neurobehavior regulation. We found that deletion of *atg7*, which is vital for autophagy, from myeloid cell-specific lysozyme M-Cre mice resulted in social behavioral defects and repetitive behaviors, characteristic features of ASDs. These mice also had increases in dendritic spines and synaptic markers and altered connectivity between brain regions, indicating defects in synaptic refinement. Synaptosome degradation was impaired in *atg7*-deficient microglia and immature dendritic filopodia were increased in neurons co-cultured with *atg7*-deficient microglia. To our knowledge, our results are the first to show the role of microglial autophagy in the regulation of the synapse and neurobehaviors. We anticipate our results to be a starting point for more comprehensive studies of microglial autophagy in ASDs and the development of putative therapeutics.

CD47 Protects Synapses from Excess Microglia-Mediated Pruning during Development.

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Microglia regulate synaptic circuit remodeling and phagocytose synaptic material in the healthy brain; however, the mechanisms directing microglia to engulf specific synapses and avoid others remain unknown. Here, we demonstrate that an innate immune signaling pathway protects synapses from inappropriate removal. The expression patterns of CD47 and its receptor, SIRP α , correlated with peak pruning in the developing retinogeniculate system, and mice lacking these proteins exhibited increased microglial engulfment of retinogeniculate inputs and reduced synapse numbers in the dorsal lateral geniculate nucleus. CD47-deficient mice also displayed increased functional pruning, as measured by electrophysiology. In addition, CD47 was found to be required for neuronal activity-mediated changes in engulfment, as microglia in CD47 knockout mice failed to display preferential engulfment of less active inputs. Taken together, these results demonstrate that CD47-SIRP α signaling prevents excess microglial phagocytosis and show that molecular brakes can be regulated by activity to protect specific inputs.

Microglia-neuron crosstalk: Signaling mechanism and control of synaptic transmission.

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The continuous crosstalk between microglia and neurons is required for microglia housekeeping functions and contributes to brain homeostasis. Through these exchanges, microglia take part in crucial brain functions, including development and plasticity. The alteration of neuron-microglia communication contributes to brain disease states with consequences, ranging from synaptic function to neuronal survival. This review focuses on the signaling pathways responsible for neuron-microglia crosstalk, highlighting their physiological roles and their alteration or specific involvement in disease. In particular, we discuss studies, establishing how these signaling allow microglial cells to control relevant physiological functions during brain development, including synaptic formation and circuit refinement. In addition, we highlight how microglia and neurons interact functionally to regulate highly dynamical synaptic functions. Microglia are able to release several signaling molecules involved in the regulation of synaptic activity and plasticity. On the other side, molecules of neuronal origin control microglial processes motility in an activity-dependent manner. Indeed, the continuous crosstalk between microglia and neurons is required for the sensing and housekeeping functions of microglia and contributes to the maintenance of brain homeostasis and, particularly, to the sculpting of neuronal connections during

development. These interactions lay on the delicate edge between physiological processes and homeostasis alteration in pathology and are themselves altered during neuroinflammation. The full description of these processes could be fundamental for understanding brain functioning in health and disease.
